

Counting Techniques



BENNETT
UNIVERSITY
TIMES OF INDIA GROUP

Bennett University

Combinations

Examples

Combination with repetition

Binomial Coefficients

Pigeonhole principle

Let us consider a set of n objects. An r -combination from the set of n objects is any selection of r objects, where the order of the objects does not matter.

Consider a set of letters $\{a, b, c\}$. Then $abc, acb, bac, bca, cab, cba$ represent different permutations but they all represent the same combination

The number of r combinations from a set of n objects is denoted by

$C(n, r)$ or nC_r

$$nC_r = \frac{n!}{r!(n-r)!}$$

In how many ways can 3 cards be selected from a pack of cards?

Solution

There are 52 cards in a pack of cards.

Therefore the number of ways to select three cards is

$${}^{52}C_3 = 52! / (3!49!) = 22100$$

In how many ways can 3 diamonds and 2 clubs be selected from a pack of cards?

Solution

There are 13 diamonds and 13 clubs in a pack of cards.

Therefore the number of ways to select are

$${}^{13}C_3 \cdot {}^{13}C_2 = 22308$$

A bag contains 4 different History books and 6 different English books. In how many ways can 3 books be selected so that there is at least 1 book on each subject?

Solution

The selection of three books with atleast 1 book of each subject can be done in the following ways:

A) There can be 1 book of history and 2 books of english.

B) There can be two books of history and 1 book of english.

Number of ways for A) $4C_1 \cdot 6C_2 = 60$

Number of ways for B) $4C_2 \cdot 6C_1 = 36$

Total number of ways $= 60 + 36 = 96$

Let us consider a set of 3 numbers $\{1, 2, 3\}$. Suppose we have to form a subset of two numbers and repetition is allowed. then we will have the following possibilities:

$\{\{1,1\}, \{1,2\}, \{1,3\}, \{2,2\}, \{2,3\}, \{3,3\}\}$

However without repetition we will have only three possibilities.

$\{\{1,2\}, \{2,3\}, \{1,3\}\}$

Theorem

Let there be n elements in a set. If repetition of elements is allowed, then the number of r -combinations is given by

$C(n+r-1, r)$ or $C(n+r-1, n-1)$

1. Find the number of ways to select three balls from a bag that contains balls of 4 different colors, if repetition is allowed.

Solution

Here $n=4, r=3$

Therefore number of combinations $= C(4+3-1, 3) = C(6, 3) = 20$

2. There are some cans of coke, Pepsi and sprite in a refrigerator. In how many ways can 5 cans be selected if repetition is allowed?

Solution

Here $n=3, r=5$

Total number of combinations $= C(3+5-1, 5)$
 $= C(7, 5) = 21$

3. How many solutions does the equation $x_1 + x_2 + x_3 = 10$ have if x_1, x_2 and x_3 are non negative integers?

Solution

The problem is similar to that of finding the number of ways to select 10 objects from a set of 3 elements, where each element can be repeated any number of times.

Here $n=3, r=10$

Therefore, total number of combinations $= C(3+10-1, 10)$
 $= 66$

4. How many solutions does the equation $x_1+x_2+x_3+x_4=30$ have if $x_1 \geq 2, x_2 \geq 4, x_3 \geq 5$ and $x_4 \geq 6$ and all are integers.

Solution

$$x_1=y_1+2$$

$$x_2=y_2+4$$

$$x_3=y_3+5$$

$$x_4=y_4+6 \text{ where } (y_i \geq 0, 1 \leq i \leq 4)$$

Now the given equation is equivalent to

$$y_1+2+y_2+4+y_3+5+y_4+6=30$$

$$y_1+y_2+y_3+y_4=13$$

$$n=4, r=13$$

Therefore total number of combinations $=C(4+13-1, 13)=560$

5. How many solutions does the equation $x_1 + x_2 + x_3 = 20$ have if $0 \leq x_1 \leq 6$, $0 \leq x_2 \leq 8$ and $0 \leq x_3 \leq 9$?

Solution

10

Let n and r be positive integers such that $r \leq n$. We have already mentioned that an r -combination from a set of n elements is denoted by nC_r .

This number is also called the binomial coefficient because the number occurs as coefficient in the expansion of powers of binomial expression such as $(x + y)^n$.

Binomial Theorem

Let x and y be variables and n be a non-negative integer, then

$$(x + y)^n = \\ = n_{C_0} x^n y^0 + n_{C_1} x^{n-1} y^1 + \dots + n_{C_{(n-1)}} x^1 y^{n-1} + n_{C_n} x^0 y^n$$

The binomial coefficient satisfy many different identities. One of the most important of these identities is the pascal's identity given as Theorem

Let n and k be positive integers with $n \geq k$, then

$$nC_{k-1} + nC_k = (n+1)C_k$$

1. Find the coefficient of x^8y^5 in the expansion of $(x + y)^{13}$

Solution

$$n=13, i=5$$

Therefore the coefficient of x^8y^5 is ${}^{13}C_5 = 13!/(8!5!) = 1287$

2. Find the coefficient of x^7y^8 in the expansion of $(x + y)^{15}$

Solution

The coefficient of x^7y^8 is ${}^{15}C_8 2^7 3^8$

Suppose we have 10 boxes and 11 balls. Each box will be 1 ball each up to 10 balls but for the 11th ball we have to choose a box from 1 to 10. In this way, at least 1 box will have more than 1 ball.

The principle can be stated as

Given n pigeonholes, the minimum number of pigeons required to be sure that at least one pigeonhole is occupied by two pigeons is $n+1$.

1. Find the number of students in a class so that 2 students were born in the same month.

Solution

pigeons=students

pigeonholes(n)=months

Thus the minimum number of students so that 2 students were born in the same month= $n+1=13$

2. How many people must be there in a group to guarantee that at-least two people have the same birthday?

Solution

Number of pigeonholes(n)=365

Therefore minimum number of people required= $(n+1)=366$

3. Students are awarded marks in a subject. The maximum marks a student can obtain is 50. How many students must be in the class to be sure that atleast 2 students get the same marks?

Generalized Pigeonhole Principle

The principle can be stated as:

Given n pigeonholes, the minimum number of pigeons required to be sure that at least one pigeonhole is occupied by $k+1$ pigeons is $kn+1$

1. Find the minimum number of students in a class to be sure that 4 of them were born in the same month.

Solution

$n=12$, $K+1=4$, Thus $k=3$

The minimum number of students are $kn+1$

$$=3 \times 12 + 1$$

$$=37$$

2. Students are awarded grades A, B ,C and D. How many students must be there in a group so that atleast 6 students get the same grade.